

Contemporary C++: *Learning Modern C++ in a Modern Way*



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Agenda 18/30

Session 18. Separate compilation and Components: An example

- Separate compilation: The stack example
- References and further readings
- Q&A

120 min (incl. Q & A)



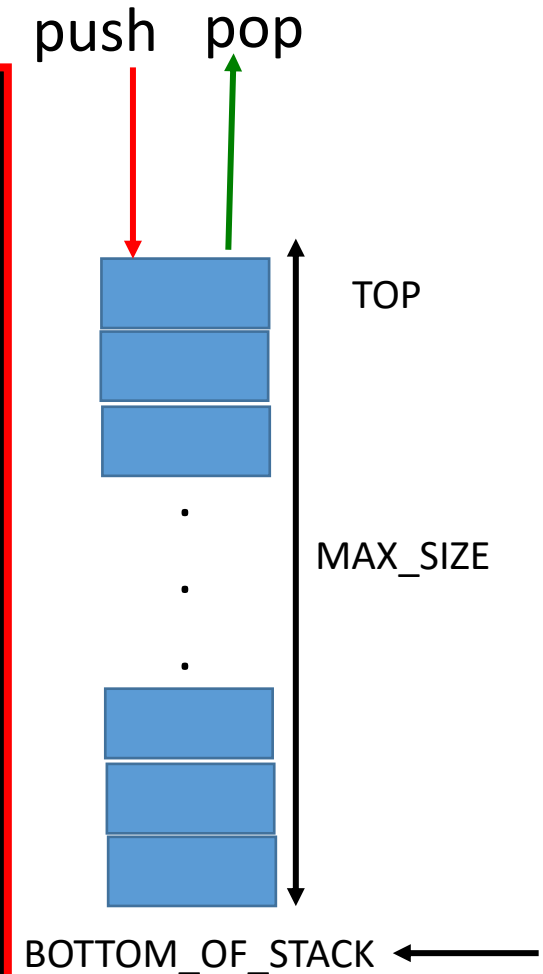
Component: An example

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- Stack interface

```
// stack.h
#include <vector>
#include <string>
/*
 * N.B: not professional but naive implementation of stack of integer
 */
const int BOTTOM_OF_STACK = -1;
const int MAX_SIZE = 10;
const int TOP = BOTTOM_OF_STACK;
const std::string stack_is_full = "pushing full Stack!";
const std::string stack_is_empty = "popping empty Stack!";
struct Stack {
    // a typical representation of stack
    std::vector<int> v;
    int top = BOTTOM_OF_STACK; // C++11: in-class member initializer
}; // of stack

void push(Stack&, int);
void pop(Stack&);
int top(Stack&);
inline bool is_full(Stack& s) { return s.top == MAX_SIZE - 1; }
inline bool is_empty(Stack& s) { return s.top == BOTTOM_OF_STACK; }
void print_stack(Stack&);
```



Component: An example

- Stack implementation

```
// stack.cpp
#include "stack.h"
#include <iostream>

void push(Stack& s, int i)
{
    if (!is_full(s)) {
        s.v.push_back(i);
        ++s.top;
    } else
        std::cout << stack_is_full << '\n';
}

void pop(Stack& s)
{
    if (!is_empty(s)) {
        s.v.pop_back();
        --s.top;
    }
    else
        std::cout << "stack is empty!" << '\n';
}
```

```
// continued
int top(Stack& s)
{
    auto t{ BOTTOM_OF_STACK };
    if (!is_empty(s)) {
        t = s.v.back();
    } else
        std::cout << stack_is_empty << '\n';

    return t;
}

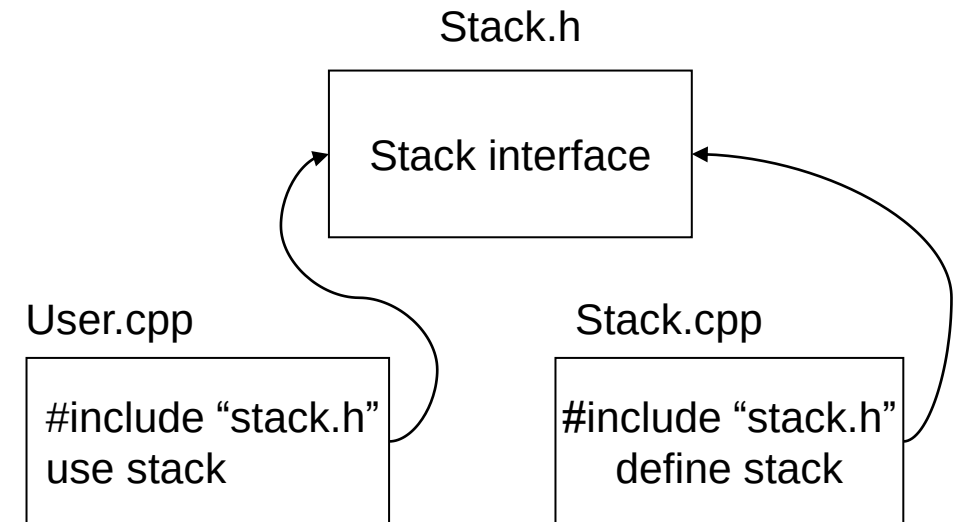
void print_stack(Stack& s)
{
    std::cout << "[ ";
    for (auto const& elem : s.v)
        std::cout << elem << ' ';
    std::cout << "]\n";
}
```

Component: An example

- Stack user

```
// stack user
#include "stack.h"
#include <iostream>
int main()
{
    Stack s;
    print_stack(s);
    pop(s);
    push(s, 42);
    push(s, -43);
    print_stack(s);
    // C++11 initializers list
    for (const auto elem : { 9, -6, 0, -3, 6, 7, 1, 1 })
        push(s, elem);
    print_stack(s);
    std::cout << "Top of stack: " << top(s) << '\n';
    push(s, 1001);

    return 0;
}
```



Thanks for your patience ...

A man who asks a question is a fool for minute,
The man who does not ask, is a fool for a life.
- Confucius

Learning to ask the right (often hard) questions is an essential part of learning to think as a programmer.

- Bjarne Stroustrup *programming Principles and Practice Using C++, page 4.*

There is no stupid question, but there is stupid answer.
- Howard Hinnant

